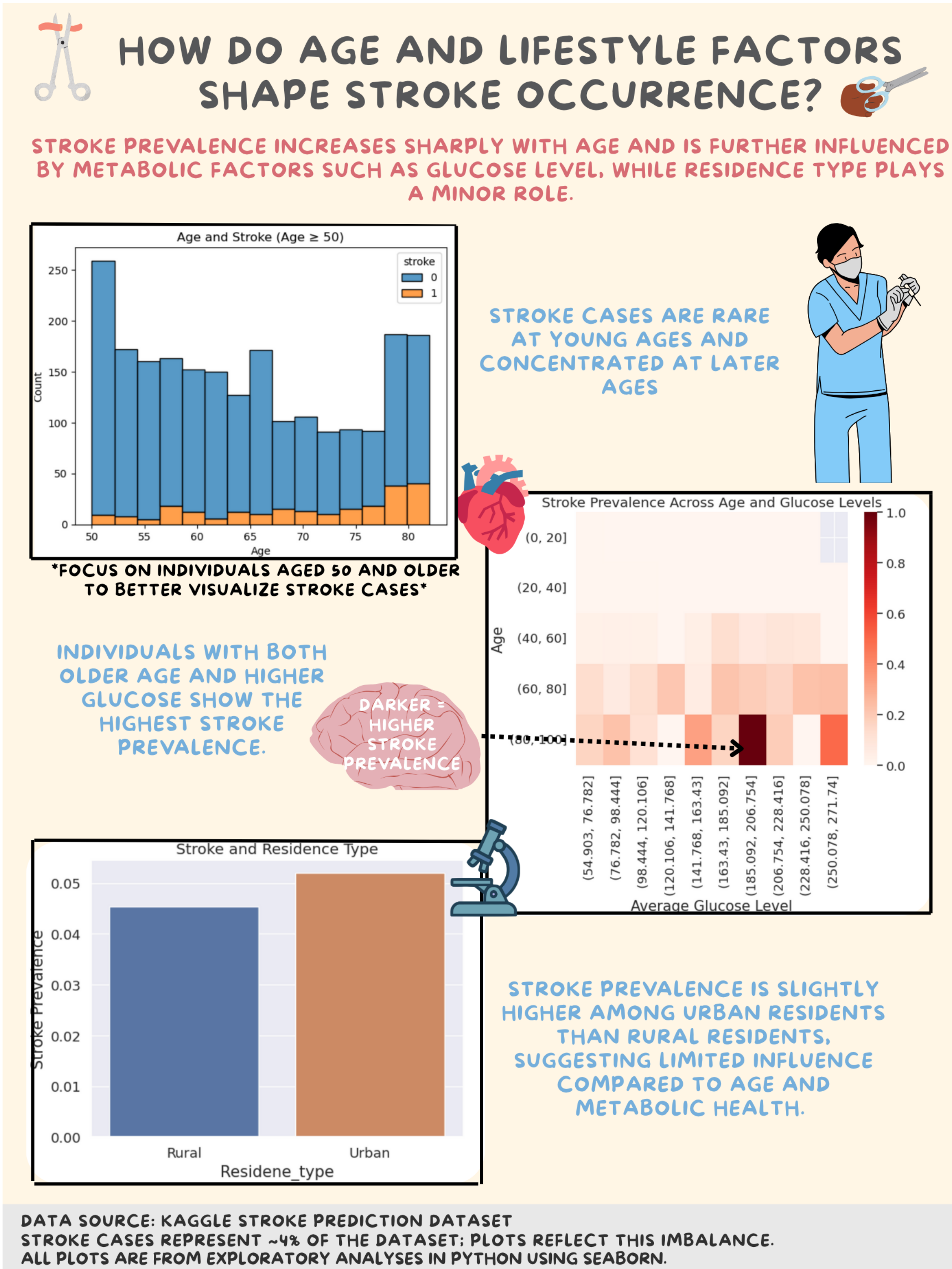
# Final_Product_#3

December 12, 2025

# 1 Final Product

# 2 How Do Age and Lifestyle Factors Shape Stroke Occurrence?



## 2.1 Description

This infographic looks at trends in stroke incidences based on demographic and lifestyle-related variables. Age was major variable that was looked at, which demonstrated the most meaningful correlation for stroke prevalence. Because stroke cases are rare at younger ages, the age histogram on the infographic is set to individuals aged 50 and older. This makes it easier to see where stroke cases actually occur and highlights that strokes are much more common at later ages.

To understand how age interacts with life style-related factors, the following infographic combines age and average glucose level into a heat map. This plot demonstrates that higher prevalence of stroke is most visible in the older age and when glucose is high, indicating that metabolic factor plays a role. In contrast, residence type is not the biggest distinguishing factor in this dataset since the bar chart comparing urban and rural populations exhibit only minor differences in stroke prevalence.

Overall, the major takeaways that emerges from this infographic are that age is the strongest factor, glucose level helps explain some of the differences within older age groups, and residence type shows minor differences. The infographic's objective was to translate these findings in a way that is easy to understand for a general audience.

## 2.2 Resources

[2]:
```python
import pandas as pd

resources = pd.DataFrame({
    "Resource Name": [
        "Stroke Dataset",
        "Notebook 1: Establishing the Data",
        "Notebook 2: Exploring the Data",
        "Notebook 3: Final Product",
        "Final Infographic (PNG)"
    ],
    "Description": [
        "Stroke dataset used for exploratory analysis",
        "Data source and description of dataset features",
        "Exploratory analysis of age, glucose, Hypertension, BMI, and residence␣
↪type",
        "Final product and project documentation/resources",
        "Infographic (PNG format)"
    ],
    "Link": [
        "https://www.kaggle.com/datasets/fedesoriano/stroke-prediction-dataset",
        "Establishing_the_Data_#1.ipynb",
        "Exploring_the_Data_#2.ipynb",
        "Final_Product_#3.ipynb",
        "Stoke_Prevalence_Infographic.png"
    ]
})
```

```
resources
```

[2]:
```
                         Resource Name  \
0                       Stroke Dataset
1  Notebook 1: Establishing the Data
2     Notebook 2: Exploring the Data
3          Notebook 3: Final Product
4            Final Infographic (PNG)

                                         Description  \
0       Stroke dataset used for exploratory analysis
1    Data source and description of dataset features
2  Exploratory analysis of age, glucose, Hyperten…
3  Final product and project documentation/resources
4                          Infographic (PNG format)

                                                Link
0  https://www.kaggle.com/datasets/fedesoriano/st…
1                     Establishing_the_Data_#1.ipynb
2                        Exploring_the_Data_#2.ipynb
3                             Final_Product_#3.ipynb
4                   Stoke_Prevalence_Infographic.png
```